

Now we can finish the proof of the theorem. If $\alpha \in C_n(X)$ is a cycle, then we have $g_{\#}(\alpha) - f_{\#}(\alpha) = \partial P(\alpha) + P\partial(\alpha) = \partial P(\alpha)$ since $\partial\alpha = 0$. Thus $g_{\#}(\alpha) - f_{\#}(\alpha)$ is a boundary, so $g_{\#}(\alpha)$ and $f_{\#}(\alpha)$ determine the same homology class, which means that g_* equals f_* on the homology class of α . $\square$

The relationship $\partial P + P\partial = g_{\#} - f_{\#}$ is expressed by saying P is a **chain homotopy** between the chain maps $f_{\#}$ and $g_{\#}$. We have just shown:

Proposition 2.12. *Chain-homotopic chain maps induce the same homomorphism on homology.* $\square$

There are also induced homomorphisms $f_* : \tilde{H}_n(X) \rightarrow \tilde{H}_n(Y)$ for reduced homology groups since $f_{\#}\varepsilon = \varepsilon f_{\#}$. The properties of induced homomorphisms we proved above hold equally well in the setting of reduced homology, with the same proofs.

Exact Sequences and Excision

It would be nice if there was always a simple relationship between the homology groups of a space X , a subspace A , and the quotient space X/A . For then one could hope to understand the homology groups of spaces such as CW complexes that can be built inductively from successively more complicated subspaces. Perhaps the simplest possible relationship would be if $H_n(X)$ contained $H_n(A)$ as a subgroup and the quotient group $H_n(X)/H_n(A)$ was isomorphic to $H_n(X/A)$. While this does hold in some cases, if it held in general then homology theory would collapse totally since every space X can be embedded as a subspace of a space with trivial homology groups, namely the cone $CX = (X \times I)/(X \times \{0\})$, which is contractible.

It turns out that this overly simple model does not have to be modified too much to get a relationship that is valid in fair generality. The novel feature of the actual relationship is that it involves the groups $H_n(X)$, $H_n(A)$, and $H_n(X/A)$ for all values of n simultaneously. In practice this is not as bad as it might sound, and in addition it has the pleasant side effect of sometimes allowing higher-dimensional homology groups to be computed in terms of lower-dimensional groups, which may already be known by induction for example.

In order to formulate the relationship we are looking for, we need an algebraic definition which is central to algebraic topology. A sequence of homomorphisms

$$\cdots \longrightarrow A_{n+1} \xrightarrow{\alpha_{n+1}} A_n \xrightarrow{\alpha_n} A_{n-1} \longrightarrow \cdots$$

is said to be **exact** if $\text{Ker } \alpha_n = \text{Im } \alpha_{n+1}$ for each n . The inclusions $\text{Im } \alpha_{n+1} \subset \text{Ker } \alpha_n$ are equivalent to $\alpha_n \alpha_{n+1} = 0$, so the sequence is a chain complex, and the opposite inclusions $\text{Ker } \alpha_n \subset \text{Im } \alpha_{n+1}$ say that the homology groups of this chain complex are trivial.

A number of basic algebraic concepts can be expressed in terms of exact sequences, for example:

- (i) $0 \rightarrow A \xrightarrow{\alpha} B$ is exact iff $\text{Ker } \alpha = 0$, i.e., α is injective.
- (ii) $A \xrightarrow{\alpha} B \rightarrow 0$ is exact iff $\text{Im } \alpha = B$, i.e., α is surjective.
- (iii) $0 \rightarrow A \xrightarrow{\alpha} B \rightarrow 0$ is exact iff α is an isomorphism, by (i) and (ii).
- (iv) $0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$ is exact iff α is injective, β is surjective, and $\text{Ker } \beta = \text{Im } \alpha$, so β induces an isomorphism $C \approx B / \text{Im } \alpha$. This can be written $C \approx B/A$ if we think of α as an inclusion of A as a subgroup of B .

An exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ as in (iv) is called a **short exact sequence**.

Exact sequences provide the right tool to relate the homology groups of a space, a subspace, and the associated quotient space:

Theorem 2.13. *If X is a space and A is a nonempty closed subspace that is a deformation retract of some neighborhood in X , then there is an exact sequence*

$$\begin{aligned} \cdots \longrightarrow \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{j_*} \tilde{H}_n(X/A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \xrightarrow{i_*} \tilde{H}_{n-1}(X) \longrightarrow \cdots \\ \cdots \longrightarrow \tilde{H}_0(X/A) \longrightarrow 0 \end{aligned}$$

where i is the inclusion $A \hookrightarrow X$ and j is the quotient map $X \rightarrow X/A$.

The map ∂ will be constructed in the course of the proof. The idea is that an element $x \in \tilde{H}_n(X/A)$ can be represented by a chain α in X with $\partial\alpha$ a cycle in A whose homology class is $\partial x \in \tilde{H}_{n-1}(A)$.

Pairs of spaces (X, A) satisfying the hypothesis of the theorem will be called **good pairs**. For example, if X is a CW complex and A is a nonempty subcomplex, then (X, A) is a good pair by Proposition A.5 in the Appendix.

Corollary 2.14. $\tilde{H}_n(S^n) \approx \mathbb{Z}$ and $\tilde{H}_i(S^n) = 0$ for $i \neq n$.

Proof: For $n > 0$ take $(X, A) = (D^n, S^{n-1})$ so $X/A = S^n$. The terms $\tilde{H}_i(D^n)$ in the long exact sequence for this pair are zero since D^n is contractible. Exactness of the sequence then implies that the maps $\tilde{H}_i(S^n) \xrightarrow{\partial} \tilde{H}_{i-1}(S^{n-1})$ are isomorphisms for $i > 0$ and that $\tilde{H}_0(S^n) = 0$. The result now follows by induction on n , starting with the case of S^0 where the result holds by Propositions 2.6 and 2.8. $\square$

As an application of this calculation we have the following classical theorem of Brouwer, the 2-dimensional case of which was proved in §1.1.

Corollary 2.15. ∂D^n is not a retract of D^n . Hence every map $f: D^n \rightarrow D^n$ has a fixed point.

Proof: If $r: D^n \rightarrow \partial D^n$ is a retraction, then $ri = 1$ for $i: \partial D^n \rightarrow D^n$ the inclusion map. The composition $\tilde{H}_{n-1}(\partial D^n) \xrightarrow{i_*} \tilde{H}_{n-1}(D^n) \xrightarrow{r_*} \tilde{H}_{n-1}(\partial D^n)$ is then the identity map

on $\tilde{H}_{n-1}(\partial D^n) \approx \mathbb{Z}$. But i_* and r_* are both 0 since $\tilde{H}_{n-1}(D^n) = 0$, and we have a contradiction. The statement about fixed points follows as in Theorem 1.9. $\square$

The derivation of the exact sequence of homology groups for a good pair (X, A) will be rather a long story. We will in fact derive a more general exact sequence which holds for arbitrary pairs (X, A) , but with the homology groups of the quotient space X/A replaced by *relative homology groups*, denoted $H_n(X, A)$. These turn out to be quite useful for many other purposes as well.

Relative Homology Groups

It sometimes happens that by ignoring a certain amount of data or structure one obtains a simpler, more flexible theory which, almost paradoxically, can give results not readily obtainable in the original setting. A familiar instance of this is arithmetic mod n , where one ignores multiples of n . Relative homology is another example. In this case what one ignores is all singular chains in a subspace of the given space.

Relative homology groups are defined in the following way. Given a space X and a subspace $A \subset X$, let $C_n(X, A)$ be the quotient group $C_n(X)/C_n(A)$. Thus chains in A are trivial in $C_n(X, A)$. Since the boundary map $\partial: C_n(X) \rightarrow C_{n-1}(X)$ takes $C_n(A)$ to $C_{n-1}(A)$, it induces a quotient boundary map $\partial: C_n(X, A) \rightarrow C_{n-1}(X, A)$. Letting n vary, we have a sequence of boundary maps

$$\cdots \rightarrow C_n(X, A) \xrightarrow{\partial} C_{n-1}(X, A) \rightarrow \cdots$$

The relation $\partial^2 = 0$ holds for these boundary maps since it holds before passing to quotient groups. So we have a chain complex, and the homology groups $\text{Ker } \partial / \text{Im } \partial$ of this chain complex are by definition the **relative homology groups** $H_n(X, A)$. By considering the definition of the relative boundary map we see:

- Elements of $H_n(X, A)$ are represented by **relative cycles**: n -chains $\alpha \in C_n(X)$ such that $\partial\alpha \in C_{n-1}(A)$.
- A relative cycle α is trivial in $H_n(X, A)$ iff it is a **relative boundary**: $\alpha = \partial\beta + \gamma$ for some $\beta \in C_{n+1}(X)$ and $\gamma \in C_n(A)$.

These properties make precise the intuitive idea that $H_n(X, A)$ is ‘homology of X modulo A .’

The quotient $C_n(X)/C_n(A)$ could also be viewed as a subgroup of $C_n(X)$, the subgroup with basis the singular n -simplices $\sigma: \Delta^n \rightarrow X$ whose image is not contained in A . However, the boundary map does not take this subgroup of $C_n(X)$ to the corresponding subgroup of $C_{n-1}(X)$, so it is usually better to regard $C_n(X, A)$ as a quotient rather than a subgroup of $C_n(X)$.

Our goal now is to show that the relative homology groups $H_n(X, A)$ for any pair (X, A) fit into a long exact sequence

$$\begin{aligned} \cdots \rightarrow H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A) \rightarrow H_{n-1}(X) \rightarrow \cdots \\ \cdots \rightarrow H_0(X, A) \rightarrow 0 \end{aligned}$$

This will be entirely a matter of algebra. To start the process, consider the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_n(A) & \xrightarrow{i} & C_n(X) & \xrightarrow{j} & C_n(X, A) \longrightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \longrightarrow & C_{n-1}(A) & \xrightarrow{i} & C_{n-1}(X) & \xrightarrow{j} & C_{n-1}(X, A) \longrightarrow 0 \end{array}$$

where i is inclusion and j is the quotient map. The diagram is commutative by the definition of the boundary maps. Letting n vary, and drawing these short exact sequences vertically rather than horizontally, we

have a large commutative diagram of the form shown at the right, where the columns are exact and the rows are chain complexes which we denote A , B , and C . Such a diagram is called a **short exact sequence of chain complexes**. We will show that when we pass to homology groups, this short exact sequence of chain complexes stretches out into a long exact sequence of homology groups

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & A_{n+1} & \xrightarrow{\partial} & A_n & \xrightarrow{\partial} & A_{n-1} \longrightarrow \cdots \\ & & \downarrow i & & \downarrow i & & \downarrow i \\ \cdots & \longrightarrow & B_{n+1} & \xrightarrow{\partial} & B_n & \xrightarrow{\partial} & B_{n-1} \longrightarrow \cdots \\ & & \downarrow j & & \downarrow j & & \downarrow j \\ \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \rightarrow \cdots$$

where $H_n(A)$ denotes the homology group $\text{Ker } \partial / \text{Im } \partial$ at A_n in the chain complex A , and $H_n(B)$ and $H_n(C)$ are defined similarly.

The commutativity of the squares in the short exact sequence of chain complexes means that i and j are chain maps. These therefore induce maps i_* and j_* on homology. To define the boundary map $\partial : H_n(C) \rightarrow H_{n-1}(A)$, let $c \in C_n$ be a cycle. Since j is onto, $c = j(b)$ for some $b \in B_n$. The element $\partial b \in B_{n-1}$ is in $\text{Ker } j$ since $j(\partial b) = \partial j(b) = \partial c = 0$. So $\partial b = i(a)$ for some $a \in A_{n-1}$ since $\text{Ker } j = \text{Im } i$. Note that $\partial a = 0$ since $i(\partial a) = \partial i(a) = \partial \partial b = 0$ and i is injective. We define $\partial : H_n(C) \rightarrow H_{n-1}(A)$ by sending the homology class of c to the homology class of a , $\partial[c] = [a]$. This is well-defined since:

$$\begin{array}{ccccc} & & a & & \\ & & \downarrow & & \\ b & \longmapsto & \partial b & \xrightarrow{i} & A_{n-1} \\ \downarrow & & \downarrow & & \downarrow \\ c & \longmapsto & \partial c & \xrightarrow{j} & B_{n-1} \\ & & \downarrow & & \\ & & C_n & & \end{array}$$

- The element a is uniquely determined by ∂b since i is injective.
- A different choice b' for b would have $j(b') = j(b)$, so $b' - b$ is in $\text{Ker } j = \text{Im } i$. Thus $b' - b = i(a')$ for some a' , hence $b' = b + i(a')$. The effect of replacing b by $b + i(a')$ is to change a to the homologous element $a + \partial a'$ since $i(a + \partial a') = i(a) + i(\partial a') = \partial b + \partial i(a') = \partial(b + i(a'))$.
- A different choice of c within its homology class would have the form $c + \partial c'$. Since $c' = j(b')$ for some b' , we then have $c + \partial c' = c + \partial j(b') = c + j(\partial b') = j(b + \partial b')$, so b is replaced by $b + \partial b'$, which leaves ∂b and therefore also a unchanged.

The map $\partial: H_n(C) \rightarrow H_{n-1}(A)$ is a homomorphism since if $\partial[c_1] = [a_1]$ and $\partial[c_2] = [a_2]$ via elements b_1 and b_2 as above, then $j(b_1 + b_2) = j(b_1) + j(b_2) = c_1 + c_2$ and $i(a_1 + a_2) = i(a_1) + i(a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$, so $\partial([c_1] + [c_2]) = [a_1] + [a_2]$.

Theorem 2.16. *The sequence of homology groups*

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \rightarrow \cdots$$

is exact.

Proof: There are six things to verify:

$\text{Im } i_* \subset \text{Ker } j_*$. This is immediate since $ji = 0$ implies $j_*i_* = 0$.

$\text{Im } j_* \subset \text{Ker } \partial$. We have $\partial j_* = 0$ since in this case $\partial b = 0$ in the definition of ∂ .

$\text{Im } \partial \subset \text{Ker } i_*$. Here $i_*\partial = 0$ since $i_*\partial$ takes $[c]$ to $[\partial b] = 0$.

$\text{Ker } j_* \subset \text{Im } i_*$. A homology class in $\text{Ker } j_*$ is represented by a cycle $b \in B_n$ with $j(b)$ a boundary, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Since j is surjective, $c' = j(b')$ for some $b' \in B_{n+1}$. We have $j(b - \partial b') = j(b) - j(\partial b') = j(b) - \partial j(b') = 0$ since $\partial j(b') = \partial c' = j(b)$. So $b - \partial b' = i(a)$ for some $a \in A_n$. This a is a cycle since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ and i is injective. Thus $i_*[a] = [b - \partial b'] = [b]$, showing that i_* maps onto $\text{Ker } j_*$.

$\text{Ker } \partial \subset \text{Im } j_*$. In the notation used in the definition of ∂ , if c represents a homology class in $\text{Ker } \partial$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle since $\partial(b - i(a')) = \partial b - \partial i(a') = \partial b - i(\partial a') = \partial b - i(a) = 0$. And $j(b - i(a')) = j(b) - ji(a') = j(b) = c$, so j_* maps $[b - i(a')]$ to $[c]$.

$\text{Ker } i_* \subset \text{Im } \partial$. Given a cycle $a \in A_{n-1}$ such that $i(a) = \partial b$ for some $b \in B_n$, then $j(b)$ is a cycle since $\partial j(b) = j(\partial b) = ji(a) = 0$, and ∂ takes $[j(b)]$ to $[a]$. $\square$

This theorem represents the beginnings of the subject of homological algebra. The method of proof is sometimes called *diagram chasing*.

Returning to topology, the preceding algebraic theorem yields a long exact sequence of homology groups:

$$\begin{aligned} \cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \rightarrow \cdots \\ \cdots \rightarrow H_0(X, A) \rightarrow 0 \end{aligned}$$

The boundary map $\partial: H_n(X, A) \rightarrow H_{n-1}(A)$ has a very simple description: If a class $[\alpha] \in H_n(X, A)$ is represented by a relative cycle α , then $\partial[\alpha]$ is the class of the cycle $\partial\alpha$ in $H_{n-1}(A)$. This is immediate from the algebraic definition of the boundary homomorphism in the long exact sequence of homology groups associated to a short exact sequence of chain complexes.

This long exact sequence makes precise the idea that the groups $H_n(X, A)$ measure the difference between the groups $H_n(X)$ and $H_n(A)$. In particular, exactness

implies that if $H_n(X, A) = 0$ for all n , then the inclusion $A \hookrightarrow X$ induces isomorphisms $H_n(A) \approx H_n(X)$ for all n , by the remark (iii) following the definition of exactness. The converse is also true according to an exercise at the end of this section.

There is a completely analogous long exact sequence of reduced homology groups for a pair (X, A) with $A \neq \emptyset$. This comes from applying the preceding algebraic machinery to the short exact sequence of chain complexes formed by the short exact sequences $0 \rightarrow C_n(A) \rightarrow C_n(X) \rightarrow C_n(X, A) \rightarrow 0$ in nonnegative dimensions, augmented by the short exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{\mathbb{1}} \mathbb{Z} \rightarrow 0 \rightarrow 0$ in dimension -1 . In particular this means that $\tilde{H}_n(X, A)$ is the same as $H_n(X, A)$ for all n , when $A \neq \emptyset$.

Example 2.17. In the long exact sequence of reduced homology groups for the pair $(D^n, \partial D^n)$, the maps $H_i(D^n, \partial D^n) \xrightarrow{\partial} \tilde{H}_{i-1}(S^{n-1})$ are isomorphisms for all $i > 0$ since the remaining terms $\tilde{H}_i(D^n)$ are zero for all i . Thus we obtain the calculation

$$H_i(D^n, \partial D^n) \approx \begin{cases} \mathbb{Z} & \text{for } i = n \\ 0 & \text{otherwise} \end{cases}$$

Example 2.18. Applying the long exact sequence of reduced homology groups to a pair (X, x_0) with $x_0 \in X$ yields isomorphisms $H_n(X, x_0) \approx \tilde{H}_n(X)$ for all n since $\tilde{H}_n(x_0) = 0$ for all n .

There are induced homomorphisms for relative homology just as there are in the nonrelative, or ‘absolute,’ case. A map $f: X \rightarrow Y$ with $f(A) \subset B$, or more concisely $f: (X, A) \rightarrow (Y, B)$, induces homomorphisms $f_\# : C_n(X, A) \rightarrow C_n(Y, B)$ since the chain map $f_\# : C_n(X) \rightarrow C_n(Y)$ takes $C_n(A)$ to $C_n(B)$, so we get a well-defined map on quotients, $f_\# : C_n(X, A) \rightarrow C_n(Y, B)$. The relation $f_\# \partial = \partial f_\#$ holds for relative chains since it holds for absolute chains. By Proposition 2.9 we then have induced homomorphisms $f_* : H_n(X, A) \rightarrow H_n(Y, B)$.

Proposition 2.19. *If two maps $f, g: (X, A) \rightarrow (Y, B)$ are homotopic through maps of pairs $(X, A) \rightarrow (Y, B)$, then $f_* = g_* : H_n(X, A) \rightarrow H_n(Y, B)$.*

Proof: The prism operator P from the proof of Theorem 2.10 takes $C_n(A)$ to $C_{n+1}(B)$, hence induces a relative prism operator $P : C_n(X, A) \rightarrow C_{n+1}(Y, B)$. Since we are just passing to quotient groups, the formula $\partial P + P \partial = g_\# - f_\#$ remains valid. Thus the maps $f_\#$ and $g_\#$ on relative chain groups are chain homotopic, and hence they induce the same homomorphism on relative homology groups. $\square$

An easy generalization of the long exact sequence of a pair (X, A) is the long exact sequence of a triple (X, A, B) , where $B \subset A \subset X$:

$$\cdots \rightarrow H_n(A, B) \rightarrow H_n(X, B) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A, B) \rightarrow \cdots$$

This is the long exact sequence of homology groups associated to the short exact sequence of chain complexes formed by the short exact sequences

$$0 \rightarrow C_n(A, B) \rightarrow C_n(X, B) \rightarrow C_n(X, A) \rightarrow 0$$

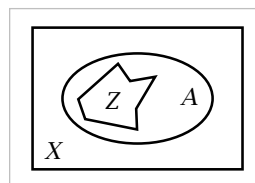
For example, taking B to be a point, the long exact sequence of the triple (X, A, B) becomes the long exact sequence of reduced homology for the pair (X, A) .

Excision

A fundamental property of relative homology groups is given by the following **Excision Theorem**, describing when the relative groups $H_n(X, A)$ are unaffected by deleting, or excising, a subset $Z \subset A$.

Theorem 2.20. *Given subspaces $Z \subset A \subset X$ such that the closure of Z is contained in the interior of A , then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces isomorphisms $H_n(X - Z, A - Z) \rightarrow H_n(X, A)$ for all n . Equivalently, for subspaces $A, B \subset X$ whose interiors cover X , the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphisms $H_n(B, A \cap B) \rightarrow H_n(X, A)$ for all n .*

The translation between the two versions is obtained by setting $B = X - Z$ and $Z = X - B$. Then $A \cap B = A - Z$ and the condition $\text{cl } Z \subset \text{int } A$ is equivalent to $X = \text{int } A \cup \text{int } B$ since $X - \text{int } B = \text{cl } Z$.



The proof of the excision theorem will involve a rather lengthy technical detour involving a construction known as barycentric subdivision, which allows homology groups to be computed using small singular simplices. In a metric space ‘smallness’ can be defined in terms of diameters, but for general spaces it will be defined in terms of covers.

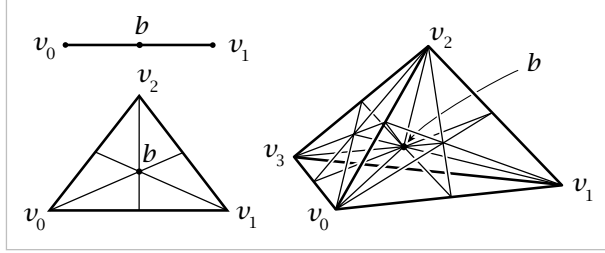
For a space X , let $\mathcal{U} = \{U_j\}$ be a collection of subspaces of X whose interiors form an open cover of X , and let $C_n^{\mathcal{U}}(X)$ be the subgroup of $C_n(X)$ consisting of chains $\sum_i n_i \sigma_i$ such that each σ_i has image contained in some set in the cover $\mathcal{U}$. The boundary map $\partial: C_n(X) \rightarrow C_{n-1}(X)$ takes $C_n^{\mathcal{U}}(X)$ to $C_{n-1}^{\mathcal{U}}(X)$, so the groups $C_n^{\mathcal{U}}(X)$ form a chain complex. We denote the homology groups of this chain complex by $H_n^{\mathcal{U}}(X)$.

Proposition 2.21. *The inclusion $\iota: C_n^{\mathcal{U}}(X) \hookrightarrow C_n(X)$ is a chain homotopy equivalence, that is, there is a chain map $\rho: C_n(X) \rightarrow C_n^{\mathcal{U}}(X)$ such that $\iota\rho$ and $\rho\iota$ are chain homotopic to the identity. Hence ι induces isomorphisms $H_n^{\mathcal{U}}(X) \approx H_n(X)$ for all n .*

Proof: The barycentric subdivision process will be performed at four levels, beginning with the most geometric and becoming increasingly algebraic.

(1) Barycentric Subdivision of Simplices. The points of a simplex $[v_0, \dots, v_n]$ are the linear combinations $\sum_i t_i v_i$ with $\sum_i t_i = 1$ and $t_i \geq 0$ for each i . The **barycenter** or ‘center of gravity’ of the simplex $[v_0, \dots, v_n]$ is the point $b = \sum_i t_i v_i$ whose barycentric coordinates t_i are all equal, namely $t_i = 1/(n+1)$ for each i . The **barycentric subdivision** of $[v_0, \dots, v_n]$ is the decomposition of $[v_0, \dots, v_n]$ into the n -simplices $[b, w_0, \dots, w_{n-1}]$ where, inductively, $[w_0, \dots, w_{n-1}]$ is an $(n-1)$ -simplex in the

barycentric subdivision of a face $[v_0, \dots, \hat{v}_i, \dots, v_n]$. The induction starts with the case $n = 0$ when the barycentric subdivision of $[v_0]$ is defined to be just $[v_0]$ itself. The next two cases $n = 1, 2$ and part of the case $n = 3$ are shown in the figure. It follows from the inductive definition that the vertices of simplices in the barycentric subdivision of $[v_0, \dots, v_n]$ are exactly the barycenters of all the k -dimensional faces $[v_{i_0}, \dots, v_{i_k}]$ of $[v_0, \dots, v_n]$ for $0 \leq k \leq n$. When $k = 0$ this gives the original vertices v_i since the barycenter of a 0-simplex is itself. The barycenter of $[v_{i_0}, \dots, v_{i_k}]$ has barycentric coordinates $t_i = 1/(k+1)$ for $i = i_0, \dots, i_k$ and $t_i = 0$ otherwise.

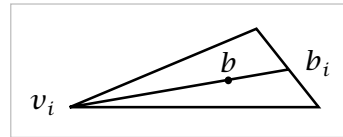


The n -simplices of the barycentric subdivision of Δ^n , together with all their faces, do in fact form a Δ -complex structure on Δ^n , indeed a simplicial complex structure, though we shall not need to know this in what follows.

A fact we will need is that the diameter of each simplex of the barycentric subdivision of $[v_0, \dots, v_n]$ is at most $n/(n+1)$ times the diameter of $[v_0, \dots, v_n]$. Here the diameter of a simplex is by definition the maximum distance between any two of its points, and we are using the metric from the ambient Euclidean space $\mathbb{R}^m$ containing $[v_0, \dots, v_n]$. The diameter of a simplex equals the maximum distance between any of its vertices because the distance between two points v and $\sum_i t_i v_i$ of $[v_0, \dots, v_n]$ satisfies the inequality

$$|v - \sum_i t_i v_i| = |\sum_i t_i (v - v_i)| \leq \sum_i t_i |v - v_i| \leq \sum_i t_i \max |v - v_i| = \max |v - v_i|$$

To obtain the bound $n/(n+1)$ on the ratio of diameters, we therefore need to verify that the distance between any two vertices w_j and w_k of a simplex $[w_0, \dots, w_n]$ of the barycentric subdivision of $[v_0, \dots, v_n]$ is at most $n/(n+1)$ times the diameter of $[v_0, \dots, v_n]$. If neither w_i nor w_j is the barycenter b of $[v_0, \dots, v_n]$, then these two points lie in a proper face of $[v_0, \dots, v_n]$ and we are done by induction on n . So we may suppose w_j , say, is the barycenter b , and then by the previous displayed inequality we may take w_k to be a vertex v_i . Let b_i be the barycenter of $[v_0, \dots, \hat{v}_i, \dots, v_n]$, with all barycentric coordinates equal to $1/n$ except for $t_i = 0$. Then we have $b = \frac{1}{n+1} v_i + \frac{n}{n+1} b_i$. The sum of the two coefficients is 1, so b lies on the line segment $[v_i, b_i]$ from v_i to b_i , and the distance from b to v_i is $n/(n+1)$ times the length of $[v_i, b_i]$. Hence the distance from b to v_i is bounded by $n/(n+1)$ times the diameter of $[v_0, \dots, v_n]$.



The significance of the factor $n/(n+1)$ is that by repeated barycentric subdivision we can produce simplices of arbitrarily small diameter since $(n/(n+1))^r$ approaches